AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM105
Introduction to Computer Programming

Final Examination
Spring Semester 2025/2026

Date: 3/6/2026

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 2 hours

Instructions:

- Total Marks: **100**
- This exam consists of **5** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Question 1 [20 Marks]

For each question, choose the most appropriate answer. Copy the answer to your answer booklet. Each question carries 2 marks.

1. Which of the following expressions evaluates to an integer value in Java?

a. `17/5`
b. `17/5.0`
c. `Math.sqrt(16)`
d. `2.5+1`

2. Which of the following is a valid Java identifier?

a. `studentMark`
b. `2student`
c. `class`
d. `student mark`

3. What is the value of `x` after executing the following statement: `int x=23%5;`

a. 3
b. 4
c. 5
d. 23

4. If `marks` is an initialized 2D array, the element in the third row and first column is:

a. `marks[3][1]`
b. `marks[2][0]`
c. `marks[0][2]`
d. `marks[1][3]`

5. Which method is used to compare the contents of two String objects?

a. `==`
b. `equals()`
c. `charAt()`
d. `length()`

6. What is the value stored in `nums[3]` after executing the following statement:
`int[] nums = new int[4];`

a. `3`
b. `4`
c. `0`
d. It is not initialized

7. When declaring a method, we use _____ to indicate that the method does not return any value.

a. `public`
b. `static`
c. `void`
d. All answers are correct

8. What will be printed after executing the following code?
`System.out.println((12 > 20) ? "High" : "Low");`

a. Low
b. High
c. True
d. False

9. A `do..while` loop executes its body at least once before checking the condition.

a. True
b. False

10. In Java, array indexes start from 1.

a. True
b. False

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Question 2 [20 Marks]

Trace the following 5 pieces of code and write their exact output [20 Marks]:

1. (3 marks)

```
for (int i = 1; i <= 3; i ++) {
    for (int j = 1; j <= 4; j++)
        System.out.print(j);
    System.out.println();
}
```

2. (3 marks)

```
int num = 0;
switch (num){
  case 0: System.out.println("ZERO");
  case 1: System.out.println("ONE");
  default:System.out.println("DEFAULT");
}
```

3. (4 marks)

```
public class FinalExam {
  public static void main(String[] args) {
     int a = 11, b = 5;
     process(a, b);
     System.out.println(a);
     System.out.println(b);
  }
  public static void process(int x, int y) {
     x--;
     y += x;
     System.out.println(x);
     System.out.println(y);
  }
}
```

4. (5 marks)

```
int[] a = {4, 2, 6, 1, 3};
a[0] = a[2] - a[4];
a[1] += a[2];
a[3] = a[4]++;
for (int i = 0; i < a.length; i++)
    System.out.println(a[i]);
```

5. (5 marks)

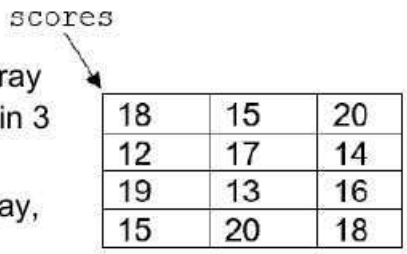
```
for (int i = 1; i <= 21; i+=3) {
    System.out.println(i);
    if (i > 10)
      break;
}
```

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Question 3 [15 Marks]

Write Java code to do the following:

1. Declare and initialize, using shorthand notation, a 2D array called `scores` that represents the scores of 4 students in 3 quizzes, as shown in the given image.
2. Using nested loops to go through all elements of the array, find and print the lowest score..



For questions 4 and 5, copy the below code into your answer booklet and continue the rest of the code according to the given specifications.

```
import java.util.Scanner;
public class FinalExam {
      public static void main(String[] args) {
          Scanner input = new Scanner(System.in);

    }
}
```

Question 4 [20 Marks]

Write Java code to do the following:

1. Ask the user to enter students' grades repeatedly (-1 to stop).
2. Each valid grade is a real number between 0 and 100 inclusive.
3. Count and print: the number of students who passed, the number of students who failed and the number of invalid grades entered.

   A student passes if the grade is greater than or equal to 50. Otherwise, the student fails.
   Invalid grades are grades less than 0 or greater than 100, except -1, which is used only to stop the input and should not be counted.

Here is a sample run to help you understand how the program works:

```
Enter the grades (-1 to stop): 25 56 77 111 90 81 -9 44 10 39 97 -1
Number of passed students: 5
Number of failed students: 4
Number of invalid grades: 2
```

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Question 5 [25 Marks]

Write Java code to do the following:

1. Write a method called `printWords()` that takes 2 arguments: an array of Strings and an integer representing a required length. The method should print all words in the array whose length is equal to the given required length. The method should not return any value.
2. In the main method, create a one-dimensional array of 8 Strings.
3. Initialize the array with 8 words entered by the user.
4. Read an integer representing the required length.
5. Call the method `printWords()` with the 2 required arguments to show the results.

Here is a sample run to help you understand how the program works:

```
Enter 8 words: Java Error Input Method Array Class Object Scanner
Enter the required length: 5
Words with length 5 are:
Error
Input
Array
Class
```

End of questions

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